ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

Project ID: 25-26J-106

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

August 2025

COLLABORATIVE TEXTURE RECOGNITION

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To all who have supported and guided me, directly or indirectly, I extend my deepest appreciation.

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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(Mr. Amila Alexander) Date: 2025/08/31

ABSTRACT

The integration of artificial intelligence, robotics, and advanced sensing technologies has opened up new possibilities for human–robot collaboration (HRC) in industrial automation. The research, which is titles Human–Robot Collaboration for Intelligent Object Sorting, will design and implement a hybrid system for a robotic arm to identify, classify, and sort objects in collaboration with humans.

The literature review covers vision-based methods, sensor-based material identification, and human–robot collaborative systems. The research gap is set in the limited accuracy of standalone vision/sensor systems and the absence of adaptive collaboration mechanisms where humans and robots work in coordination in dynamic sorting environments.

The proposed solution combines computer vision (object detection and image recognition models), sensor fusion (vision, capacitive, and ultrasonic sensors), and an HRC framework with safety measures for high-precision object recognition and sorting. Effective communication protocols among robots and human operators will be further investigated to ensure safety, flexibility, and efficiency.

Expected results include a prototype capable of real-time material recognition (metal, plastic, rubber, glass), accurate sorting, and safe co-working with human laborers. The work will be useful to smart manufacturing, recycling systems, and warehouse automation, with both research and industrial values.

Keywords: Human–Robot Collaboration, Computer Vision, Sensor-Based Identification, Intelligent Sorting, Industrial Robotics

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LIST OF ABBREVIATIONS

Abbreviation	Description
HRC	Human Robot Collaboration
AI	Artificial Intelligence
CNN	Convolutional Neural Network
ResNet	Residual Network
RL	Reinforcement Learning
NLP	Natural Language Processing
XAI	Explainable Artificial Intelligence
ML	Machine Learning
HRI	Human Robot Interaction
BERT	Bidirectional Encoder Representations from Transformers
PCA	Principal Component Analysis
YOLO	You Only Look Once

1. INTRODUCTION

1.1. Background & Literature survey

1.1.1. Background

Robotic sorting systems have become a vital part of modern automation, enabling industries to handle a wide range of materials efficiently and accurately. Their applications span from manufacturing and agriculture to recycling and logistics, where the ability to classify and separate objects with speed and precision is essential for productivity. However, despite significant advancements in robotics and artificial intelligence (AI), real-world deployments of sorting robots still face numerous challenges. These challenges are particularly evident in environments characterized by diverse textures, dynamic human interactions, and varying operational conditions.

One of the primary technical challenges in robotic sorting is accurate material recognition when faced with varied surface textures and environmental variability. Traditional machine vision systems often struggle to maintain consistent classification performance under changing lighting conditions, background noise, or with objects of irregular shapes and surfaces [1], [2]. These limitations are especially problematic in industrial and waste-sorting applications where materials are not standardized and may exhibit considerable wear, contamination, or deformation. Conventional feature-extraction methods tend to fail under these conditions, highlighting the need for more robust and adaptive approaches.

Another significant obstacle lies in the adaptability of robotic sorting systems to dynamic and unstructured environments. In many real-world scenarios, objects arrive in unpredictable orientations, overlapping with others or partially occluded, which complicates both detection and classification tasks. Furthermore, environmental factors such as dust, moisture, or mechanical vibrations can introduce additional noise into sensor readings, reducing system reliability. These challenges necessitate the integration of advanced sensing modalities—such as multispectral imaging, depth sensing, and tactile

feedback—with AI-driven decision-making models that can adapt to such variability in real time.

In addition, human–robot interaction presents an important consideration for industrial sorting systems, especially in collaborative workspaces where robots operate alongside human workers. Robotic systems must not only identify and sort objects accurately but also anticipate and adapt to changes in human workflows, safety requirements, and operational priorities. Traditional static algorithms are insufficient for these contexts, as they cannot adjust to fluctuating task demands or unexpected operator interventions. Incorporating adaptive behavior modeling, multimodal communication channels, and explainable AI frameworks can enhance trust, transparency, and overall system efficiency in such collaborative environments.

1.1.1.1. Collaborative Texture Recognition

Recent research suggests that deep learning techniques, particularly Convolutional Neural Networks (CNNs), offer a powerful solution for texture recognition in such environments [3]. CNNs are capable of learning hierarchical representations of visual and sensor-based inputs, enabling them to identify subtle texture differences that might be overlooked by traditional algorithms. To further enhance performance and reduce the requirement for extensive training data, transfer learning with pre-trained models such as ResNet can be employed. Transfer learning allows the system to leverage prior knowledge from large-scale image datasets and adapt it to domain-specific sorting tasks [1]. Additionally, our approach integrates sensor-driven machine learning, combining visual inputs with other sensing modalities such as tactile and proximity sensors for improved recognition accuracy.

To address uncertain classification cases, the system incorporates human-in-the-loop feedback mechanisms. When the robot encounters ambiguous objects, it actively seeks input from human operators. This interaction not only improves immediate decision-making but also supports incremental learning, allowing the system to continuously refine

its classification models over time. Furthermore, when labeled data is limited, Reinforcement Learning (RL) is applied to optimize sorting policies. RL enables the robot to learn effective decision-making strategies through trial-and-error interactions with the environment, gradually improving its accuracy even in data-scarce scenarios [5].

1.1.1.2. Human-Robot Collaboration and Adaptability

Beyond technical recognition capabilities, the efficiency of sorting systems depends heavily on the adaptability of robots to human behaviors. In many real-world applications, robots and humans work side-by-side, with humans often adjusting their strategies based on task requirements, time constraints, or personal preferences. Traditional robotic systems are typically static, operating according to pre-defined rules, and are therefore unable to adjust to changes in human workflow [4].

Our proposed system addresses this by monitoring human behavior through visual and interaction data, using clustering and change detection algorithms to detect shifts in sorting strategies [6]. For example, if a human operator begins grouping objects by a newly prioritized feature, the robot dynamically updates its sorting logic to complement the human's approach. This adaptability ensures a fluid division of labor, where the robot and human can work simultaneously without interfering with each other's tasks. Such adaptability is essential for collaborative robotics (cobots), where the goal is not to replace human labor but to augment it, combining human flexibility with robotic precision [6].

The integration of behavior-adaptive algorithms also improves task allocation efficiency. By reallocating sorting responsibilities based on real-time observations, the system reduces bottlenecks, minimizes idle time, and improves overall throughput. This human-centered approach makes the robot a responsive partner rather than a rigid machine.

1.1.1.3. Multimodal Communication and Interaction

Effective communication remains a bottleneck in human-robot collaboration. In complex sorting environments, misinterpretation of commands or intentions can lead to inefficiencies or errors. Our framework addresses this by integrating multimodal sensors, including gaze tracking [7], microphones, and visual recognition systems. Gaze tracking enables the robot to infer the operator's attention focus, while microphones capture verbal instructions, which are processed using Natural Language Processing (NLP) models such as BERT [8].

This multimodal fusion allows the robot to interpret both visual and linguistic cues simultaneously, significantly reducing ambiguity in communication. For example, if a human says "sort these over there" while looking at a pile of objects, the robot uses gaze direction to determine "these" and spatial mapping to identify "over there." Adaptive dialogue management further refines communication by clarifying uncertain instructions through targeted questions, ensuring that the robot's understanding aligns with the human's intent.

The result is a more intuitive interaction between humans and robots, enabling seamless coordination even in fast-paced or noisy environments. This enhances not only task efficiency but also operator satisfaction and trust.

1.1.1.4. Explainable AI for Transparency and Trust

While technical performance is crucial, transparency in robotic decision-making is equally important for building trust in human-robot collaboration. When a robot's actions appear opaque or arbitrary, human operators may hesitate to rely on it, reducing overall system effectiveness [9]. To address this, our system integrates Explainable AI (XAI) approaches that combine interpretable machine learning models with rule-based heuristics.

The robot is capable of generating natural language explanations for its actions, supplemented by visual or voice cues to provide additional clarity. For instance, when

sorting an object, the robot might explain, “I classified this as plastic due to texture patterns detected by the sensor and confirmed by weight measurement.” Additionally, the system supports interactive queries, enabling users to ask why certain actions were taken and receive context-aware explanations [10].

This level of transparency fosters mutual understanding and trust, which is essential for safe and efficient collaboration. By aligning robotic decisions with human expectations and providing clear rationales, XAI bridges the gap between human intuition and robotic precision.

1.1.1.5. Integrated System Overview

The proposed system represents a comprehensive solution to the major challenges faced by robotic sorting in real-world, human-centered environments. By combining advanced material recognition, human-adaptive behavior modeling, multimodal communication, and explainable AI, we create a robot that is not only technically capable but also socially intelligent.

The system’s continuous learning framework ensures that it evolves alongside its human partners, adapting to changing strategies, improving its classification models, and refining communication protocols. This dynamic interplay between human and robot fosters higher efficiency, reduces sorting errors, and promotes a more natural teamwork experience.

Ultimately, our approach moves beyond the paradigm of robots as isolated tools toward a vision of true cooperative automation—where humans and robots collaborate as equal partners to achieve shared goals efficiently, safely, and transparently. By addressing the limitations of existing systems and incorporating the strengths of both human adaptability and robotic precision, we aim to unlock new potential for complex sorting tasks across industrial, agricultural, and service domains. Our objective is not to replace human workers, but to create a synergistic framework in which both humans and robots can

coexist and thrive—leveraging their respective strengths to enhance productivity, well-being, and long-term sustainability.

1.1.2. Literature survey

1.1.2.1. Collaborative Texture Recognition

Cooperative texture recognition enables robots to detect surface types of objects such as metal, plastic, glass, and rubber based on vision and sensor combined. Traditional approaches utilized vision exclusively, but visual similarity between materials (shiny metal vs. glass) and environmental variations (lighting, occlusion) generally constrain accuracy. Emerging research targets multi-modal perception, where tactile sensors and proximity sensors complement cameras to ensure accurate classification [3], [4].

For example, Mudalige et al. [3] demonstrated texture perception using CNNs on a quadruped robot via tactile sensing feet and Mao et al. [4] fused vision and tactile information for improved dexterous manipulation recognition. Ma et al. [5] extended this to multi-modal bionic fingers with a CBAM-CNN, demonstrating the power in fusing tactile and vision data. These studies highlight the need for human-robot collaboration sensing, where the robots can learn a strategy for recognition and involve humans in case of uncertainty [12], [13].

1.1.2.2. Sensor Integration for Material Classification

Robotic vision improves with combining multiple sensors. Optical and RGB cameras see patterns of texture and surface reflectance, and capacitive proximity sensors detect material-specific responses such as metallic versus non-metallic objects detection. Optical sensors are particularly useful in glass detection by their transparency and reflectivity signatures, and inductive sensors provide good response for metal surfaces [6].

Recent work focuses on the need for sensor fusion algorithms that provide a combination of information from diverse modalities. Cao et al. [7] presented a zero-shot learning platform that fused tactile and semantic data to identify new textures. Luo et al. [9] highlighted how multi-sensor integration with reinforcement learning allows robots to

learn adaptive policies under dynamic conditions. These findings validate the strategy of fusing inductive, proximity, and optical sensors with visual recognition using CNN.

1.1.2.3. Deep Learning, Transfer Learning, and Reinforcement Learning Strategies

Convolutional Neural Networks (CNNs) have widespread use in surface identification from their ability to pick out precise features from sensor and visual inputs. Transfer learning, where vision-trained CNNs are reused for use in tactile or material identification, reduces dependence on large touch datasets [11]. Ma et al. [11] demonstrated how transfer learning enables robots to generalize across material classes with little training data.

In addition, Reinforcement Learning (RL) enables robots to actively probe objects, selecting the best strategy for classification. Zhang et al. [10] utilized RL in adaptive sorting of objects, while Mao et al. [13] utilized tactile-visual RL exploration for improving recognition in collaborative sorting robots. Adaptive methods like these are necessary for handling unknown textures and dynamically adjusting recognition models based on human input.

1.1.2.4. Human-in-the-Loop and Explainable AI (XAI)

Human-robot collaboration requires human-robot interaction (HRI) to resolve ambiguities and increase system adaptability. Human-in-the-loop systems allow robots to seek human feedback when material classification confidence is low, and adaptive learning becomes easier [12].

Further, Explainable AI (XAI) methods such as Grad-CAM [14] and SHAP [15] provide understandable feedback in the form of visual or sensor features that influenced the classification. Transparency increases human trust, offers safety in industrial applications, and accelerates operator training.

1.1.2.5. Material and Surface Recognition Applications

Collaborative texture recognition finds broad applications:

- Industrial Sorting – Material identification and sorting (metal, glass, plastic, rubber) in the manufacturing or recycling process [1], [8].
- Recycling Plants – Plastic and metal separation where optical sensors cannot single-handedly because of surface likeness [3], [6].
- Food Processing – Ripeness and contamination detection on food surfaces using vision and touch sensors [2].
- Assistive Robotics – Discriminating between domestic objects (e.g., delicate glass vs. robust plastic) in order to safely manipulate them [5], [12].

1.1.2.6. Challenges in Collaborative Texture Recognition

Despite significant progress, Collaborative Texture Recognition still faces some challenges:

- Environmental Variability: Optical and proximity sensors are susceptible to light, reflection, humidity, and electromagnetic interference, which can deceive visual and tactile measurements.
- Data Scarsity: Labeled multimodal texture datasets (visual + proximity) are expensive to obtain, and current datasets remain small in size and limited in diversity, limiting the generalization of supervised deep learning models.
- Real-Time Processing: Deep CNN-based classification and multimodal integration are computationally intensive, making them difficult to deploy on real-time, edge-based robotic systems.
- Explainability: Deep learning models are typically black boxes, and they do not offer much information as to why a given material (e.g., plastic vs. metal) was categorized as such, inhibiting trust from being established in safety-critical or industrial settings.

- Adaptive Exploration: Humans modify their exploration strategies when perceiving novel texture, but robots apply static sensing that limits adaptability to new or novel materials [13].

Addressing these challenges entails combining real-time multimodal perception, zero-shot or transfer learning for novel materials, reinforcement learning for exploration that is adaptive, and human-in-the-loop collaboration to offer transparency and robustness.

1.2. Research gap

Texture recognition in robot systems has made significant leaps with the use of computer vision, tactile sensing, and multimodal deep learning methods. Traditional object sorting approaches have been primarily reliant on computer vision techniques, such as convolutional neural networks (CNNs) and YOLO-based detection models, to identify and classify objects [1], [2], [8]. While these vision-based methods succeed in well-controlled environments, they do not work in real-world image scenes where surface finishes such as metal, glass, rubber, and plastic need to be separated under various environmental conditions such as low light, glint, occlusion, or soiling [3].

It has been pointed out by recent research that tactile sensing gives complementary information to vision, enabling systems to sense surface roughness, softness, and compliance. Research such as DogTouch [3] and the multi-modal tactile-visual approaches of Mao et al. [4] establish that surface texture recognition is dramatically influenced by tactile feedback. Ma et al. [5] have proposed a multi-modal bionic finger shape and Rostamian et al. [6] demonstrated how artificial tactile and proprioceptive receptors are implemented. But a major problem is that tactile data sets are still small and expensive to acquire [5], [7], so big-data supervised training is not possible. This therefore falls back on transfer learning [11] and zero-shot learning methods [7], which are yet in their infancy of use in tactile-visual recognition.

Another fundamental gap is real-time execution. The majority of cutting-edge multimodal architectures require heavy computation for feature fusion and deep neural networks,

preventing their use in real-time robotic systems where decisions must be rendered within milliseconds [9]. Reinforcement learning techniques for sensor-based exploration [9], [13] are promising in adaptive policy, yet remain largely limited to simulation environments or controlled laboratory tests, as opposed to efficient application in noisy industrial environments.

Moreover, explainability of multimodal recognition systems is also a nascent discipline. Most multimodal recognition models are black boxes, and it is difficult for human colleagues to trust or certify robot decisions. While interpretability techniques such as Grad-CAM [14] and SHAP [15] have been proposed for machine learning and computer vision, these have not yet been thoroughly integrated into tactile-visual fusion systems. This is one of the key challenges of human–robot collaboration (HRC), where operators ought to understand and trust the robot's reasoning process [10], [12].

Finally, adaptive exploration techniques for texture recognition remain underexplored. Existing systems predominantly utilize pre-specified tactile or vision-based scanning methods, which fail to meet dynamic conditions or novel materials [12], [13]. Within human-in-the-loop strategies, Cao et al. [12] demonstrated the viability of using human guidance to enhance tactile object recognition, but this remains a new area. Most multimodal recognition systems are still devoid of robust mechanisms for real-time adaptation when faced with novel or degraded textures.

Briefly, the research gap is in the lack of a multimodal, collaborative architecture combining vision, tactile sensing, transfer learning and reinforcement learning to enable real-time, adaptive, and robust texture recognition. While independent advances are being made in tactile sensing, transfer learning, and reinforcement learning, no collaborative framework addresses the joint challenges of environmental change, sparsity of data, real-time computation, explainability, and adaptive exploration in robotic texture recognition.

Table 1.1: Research gap representation 1

Feature	IoT-based YOLO [1]	Deep RL Sorting [8]	Multi-Object DL Sorting [4]	Object Sorting Robotic Arm[5]	Proposed
Real-time classification	Limited	✓	✓	Limited	✓
Multi-sensor collaboration	✗	Limited	✗	Limited	✓
Multi-material adaptability (rubber, plastic, metal, glass)	✗	✗	✗	✗	✓
Reinforcement Learning (adaptive behavior)	✗	✓	✗	✗	✓
Hybrid Rule-based + Interpretable ML	✗	✗	✗	✗	✓

1.3. Research Problem

Despite advances in multimodal perception and robot sensing, the collaborative texture recognition issue in practical object sorting and handling remains to be addressed. Robots in industrial, medical, and service settings are typically programmed to recognize and separate objects made of materials such as metal, glass, plastic, and rubber, where the surface characteristics would be critical in effective sorting, safe manipulation, or assembly. Vision-only systems [1], [2], [8] currently fall short in such cases since they are vulnerable to environmental factors like reflections, varying illumination, and occlusion. Although multimodal tactile-visual systems [3]–[7] have proven that combining vision with touch makes the recognition more accurate, the systems also suffer from major shortcomings in terms of availability of datasets, non-scalability, and high computational burden.

Large-scale tactile data scarcity is a strong bottleneck [5], [7]. Unlike visual datasets (such as ImageNet), tactile information collection requires expert equipment, human labeling, and routine material handling, making it expensive. This severely inhibits the ability of supervised learning models to generalize over diverse textures. Transfer learning from vision-trained CNNs [11] and zero-shot learning methods [7] are limited solutions because their ability to generalize over diverse real-world materials has not been extensively tested.

Additionally, real-time processing constraint precludes practical use. Multi-modal fusion models commonly rely on high-dimensional feature extraction from both tactile and visual modalities, followed by the ensuing deep neural inference [4], [6] that is computationally costly. In industrial sorting or in real-time HRC scenarios, identification must be accomplished within milliseconds; otherwise, task execution would be interrupted. Reinforcement learning strategies for decision and exploration [9], [13] demonstrate adaptive potential, but their incorporation in multimodal recognition remains unknown territory.

Another unresolved issue is the non-explainability of collaborative texture recognition systems. The majority of current models are opaque black boxes, with no transparency regarding decision-making [14], [15]. In human–robot collaboration applications, this undermines operator trust and restricts system adoption. As stressed by Angleraud et al. [10], industrial HRC uses must be made transparent, and Cao et al. [12] demonstrated human-in-the-loop interaction enhances system flexibility. Nevertheless, none of the solutions has used XAI frameworks in tactile-visual recognition pipelines to provide intuitive, real-time human feedback.

Lastly, adaptive exploration of new materials is unexplored. Robots typically encounter new or rusted surfaces that were not encountered during learning. Current approaches are likely to have static exploration policies, limiting the robot from learning and adapting online. Although Mao et al. [13] proposed adaptive tactile-visual exploration, it is still

largely investigated for experimental environments rather than for general robotic platforms.

The research problem, therefore, is to create a Collaborative Texture Recognition System that integrates real-time multimodal perception, explainable machine learning, and interactive human-in-the-loop feedback. The system needs not only to provide accurate classification of dense textures but also to provide adaptive sorting policies and understandable reasoning to foster trust, improve efficiency, and augment human-robot collaboration in uncertain environments.

2. OBJECTIVES

2.1. Main Objective

The primary objective of this research is to design and develop a smart robotic sorting system that goes beyond traditional, rule-based automation by exhibiting adaptability, situational awareness, and effective human–robot collaboration to identify metal, rubber, plastic and glass. Unlike static sorting systems that operate on predefined parameters, the proposed system will be capable of perceiving and responding to variations in the working environment, such as changes in material properties, lighting, or object arrangement, as well as adjusting its operations in response to human actions and strategies. By integrating advanced machine learning algorithms for material recognition, behavioral analysis for understanding human interaction patterns,and transparent decision-making, the system aims to deliver an intelligent, flexible, and trustworthy solution. This will not only enhance the accuracy and efficiency of sorting operations but also foster a collaborative partnership between humans and robots, where each can dynamically complement the other's strengths in complex, real-world industrial and service environments.

2.2. Sub-Objectives

1. Data Acquisition and Labeling
    - Collect high-resolution visual and sensor data (capacitive proximity, and camera) of a diverse material set.
    - Label each sample with relevant surface properties such as material type, roughness, transparency, reflectivity, hardness, and compliance.
    - Organize an amenable dataset for multi-modal learning.
2. Feature Extraction and Preprocessing
    - Extract multi-modal features such as texture roughness, metallicity, transparency, glossiness, color, shape, and compliance.
    - Normalize and preprocess data to combat variability in lighting, sensor noise, and object orientation.

- Fuse visual and sensor-based features into unified input vectors for model training.

3. Deep Learning Model Development
    - Create a Convolutional Neural Network (CNN) for texture classification through transfer learning from pre-trained vision models.
    - Fine-tune the CNN on multi-modal data to enhance classification accuracy for ambiguous or visually similar textures.
    - Integrate attention mechanisms to focus on the most informative areas for/regions.
4. Reinforcement Learning for Adaptive Exploration
    - Use Reinforcement Learning (RL) to optimize robot exploration and object manipulation strategies as a function of recognition confidence and sensorimotor feedback.
    - Enable the robot to decide when to request human input in ambiguous or uncertain cases.
    - Improve sorting efficiency by learning adaptive strategies for real-time, collaborative environments.
5. Human-Robot Collaboration Integration
    - Combine human-in-the-loop feedback for disambiguating classification.
    - Facilitate smooth collaboration in real-world environments with humans having the ability to dynamically guide the sorting process as needed.
6. Deployment and Validation
    - Deploy trained model and RL method on robotic hardware with multi-modal sensors.
    - Test real-time texture recognition performance on varying lighting conditions, material types, and object arrangements.

3. METHODOLOGY

3.1. System Design Architecture

3.1.1. Overall Collaborative Object Sorting System Architecture

The system architecture demonstrates how four fundamental modules named Collaborative Texture Recognition and Adaptive Pattern Recognition via Human Behavior and Human-Led Sorting with Conversational Robotic Assistance and Explainable AI for Collaborative Decision-Making function together to achieve efficient object recognition and adaptive task allocation and multimodal interaction and transparent decision-making. The integrated framework enables a continuous human–robot learning cycle which promotes flexible operation and trust-based collaboration during sorting tasks.

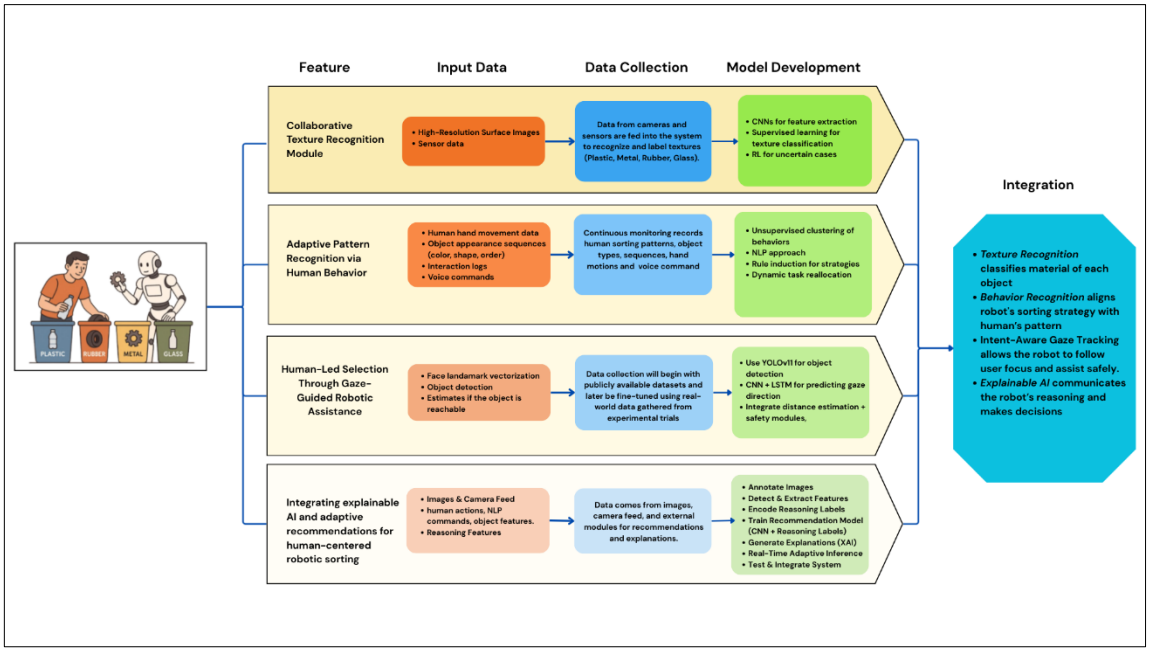

Figure 3.1: Overall System Architecture 1

3.1.2. Collaborative Texture Recognition System Design Architecture

The proposed collaborative texture recognition system on identifying and classifying object surfaces such as metal, glass, plastic, and rubber. It integrates robotic sensors and cameras to capture high-definition surface photos, which are passed through convolutional neural networks (CNNs) for feature extraction. Machine learning algorithms such as reinforcement learning refine recognition performance through human feedback, thus providing accurate and adaptive texture classifying within collaborative sorting procedures.

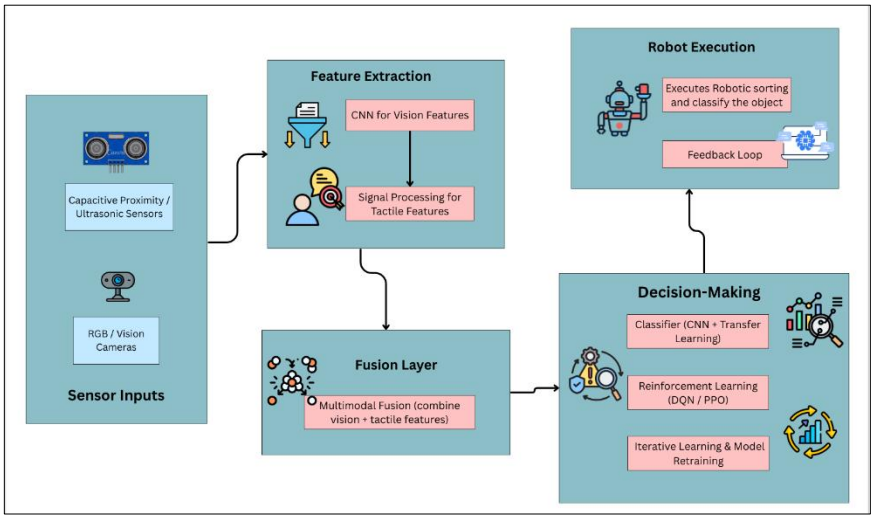

Figure 3.2: Texture recognition diagram 1

3.2. Data Acquisition

Data for this work is collected by a combination of optical and inductive/proximity sensors. The primary sources are:

- Visual Data: Captured through a high-definition RCD webcam or some other optical camera. Various images of each object are captured under various light conditions and orientations to render it resilient.
- Capacitive Proximity Data: Measures relative distance to object surfaces and detects metallic features.

The dataset contains a number of object classes: metal, glass, plastic, and rubber, with 100–500 samples per class depending on object diversity. Each sample has synchronized multi-modal sensor readings, accompanied by a timestamp for correct alignment.

3.3. Data Preprocessing

Preprocessing data ensures consistency, noise elimination, and quality improvement in features:

1. Visual Data:
    - Resize images to be of the same input size (e.g., 224×224 pixels) to align them with CNN.
    - Normalize to normalize pixel values between 0 and 1.
    - Data augmentation: rotation, flipping, brightness adjustments, and Gaussian noise addition for better generalization.
2. Capacitive Proximity Data:
    - Smooth out sensor noise using smoothing filters (e.g., moving average).
    - Normalize distance and signal strength measurements to a common scale.
3. Feature Fusion:
    - Merge visual embeddings with sensor measurements into one feature vector.
    - Reduce missing or inconsistent readings through interpolation or outliers elimination.
4. Labeling:
    - Manual object class labeling for supervised learning.
    - Human confidence scores for feedback are cached for updates in reinforcement learning.

3.4. Machine Learning Models

The system uses a hybrid learning framework:

1. Convolutional Neural Networks (CNNs):
   - Fine-tune pre-trained networks such as ResNet50 or MobileNet with collected object images.
   - Extract hierarchical visual features for texture classification.
2. Reinforcement Learning (RL):
   - Perform adaptive exploration strategies for new or unclear textures.
   - States: sensor readings, image embeddings; Actions: probing with proximity sensor or reorienting object.
   - Rewards: Correct classification, knowledge acquired, or confidence boosted.
3. Multimodal Fusion Model:
   - Combines visual and capacitive proximity features using a fully connected layer or concatenation-based fusion.
   - Provides robust classification even in missing or noisy modalities.

3.5. Tools and Technologies

3.5.1. Hardware

- Robotic Arm: Used for autonomous object manipulation, sorting, and exploration. Allows integration with multiple sensors and real-time testing of recognition models.

- Capacitive Proximity Sensors: Detect the presence and proximity of metallic or conductive materials; used to capture surface-specific properties.

- Optical Sensors: Capture surface reflectivity, transparency, and other optical features that help distinguish between materials like glass, plastic, or rubber.

- RCD Webcam / High-Resolution Camera: Provides RGB (and possibly RGB-D) images for visual texture classification using CNNs.

3.5.2. Software Tools

- Python: Main programming language for implementing machine learning, deep learning, and data preprocessing pipelines.
- TensorFlow / PyTorch: Deep learning frameworks for building CNNs, transfer learning models, and reinforcement learning agents.
- OpenCV: Image processing and feature extraction library for computer vision tasks like texture segmentation and data augmentation.
- ROS (Robot Operating System): Middleware for robotic control, sensor integration, and communication between modules.

3.6. Data Refinement

Refinement improves model accuracy and reliability:

- Outlier Removal: Sensor readings outside expected ranges are eliminated.
- Noise Reduction: Filtering of proximity and inductive signals by Gaussian or median filtering.
- Balancing Dataset: Oversampling or synthetic creation (SMOTE, data augmentation) of minority classes.
- Feature Selection: Principal Component Analysis (PCA) or correlation analysis reduces dimension and eliminates redundant features.

3.7. Evaluation and Testing

The system is evaluated by quantitative scores and real-world testing:

1. Classification Metrics:
    - Accuracy, Precision, Recall, F1-score by object class.
    - Analysis Report: Confusion matrix to identify patterns of misclassifications.
2. Adaptive Performance:
    - Evaluating how well the system can learn from unseen textures through reinforcement learning updates.
    - Human-in-the-loop interaction logged and analyzed to enhance learning.
3. Real-Time Performance:
    - Frame processing rate (FPS) and decision-making delay.
    - Evaluating if sensor fusion meets real-time constraints.
4. Robustness Testing:
    - Lighting variations, angle variations, occlusion, and distance.
    - Performance when there is noisy or missing sensor data.

The proposed methodology makes use of inductive/proximity sensors, optical sensors, and a webcam to obtain multi-modal object features. Preprocessing, feature extraction, and fusion are achieved using transfer learning and reinforcement learning models to classify object surfaces (metal, glass, plastic, rubber). Trust and adaptability are enabled with human-in-the-loop feedback and explainability techniques. Evaluation is performed with respect to accuracy, adaptive exploration, and real-time performance for effective deployment in collaborative robotics applications.

4. DESCRIPTION OF PERSONAL AND FACILITIES

4.1. Research and Development Team

The Research and Development Team will create, execute, and evaluate the proposed Explainable AI-based Recommendation System. The team combines expertise from different fields, including,

- Object recognition and sensor integration for accurate perception of size, shape, and weight.
- Human-robot collaboration modules, e.g., NLP for interactive communication.
- Recommendation system creation for real-time object sorting.
- Development of an explainable AI model for explainable robotic decision-making.

4.2. Robotics Specialists and Technical Support

A robotics specialist has been identified to assist in designing, integrating, and maintaining the robotic hardware and sensors. Their expertise ensures safe operation, proper placement and deployment of sensors, and smooth integration of the Recommendation System with the XAI modules. Additionally, the specialist provides guidance on technical challenges that may arise during implementation.

4.3. Academic Community and Supervisors

The research will also be guided and evaluated by the scientific community, including the supervisor and co-supervisor. Through their intervention, there will be ensured scientific validity, methodological quality, and compliance with research best practices. The research outputs can also provide relevant knowledge to consider for future works on robotics, human-robot interaction, and Explainable AI.

4.4. Beneficiaries

Upon successful implementation, the proposed system will have a direct impact on.

- Industrial workers in various regions will experience enhanced efficiency and reliability in object sorting processes, reduced manual labor, and fewer errors.

5. BUDGET AND BUDGET JUSTIFICATION

The proposed budget contains specific allocations which will enable the successful deployment of the multimodal human–robot communication framework. The robotic arm receives the largest funding because it performs sorting operations and proves the system's effectiveness in actual use. The system uses inductive/proximity sensors and ultrasonic sensors and safety sensors to achieve accurate environmental perception and interaction capabilities. The system uses webcams for vision-based object recognition and a dedicated USB microphone to capture high-quality speech for speech recognition operations. The system's ability to adapt to changing conditions depends on supporting devices which include a decibel meter and noise sensor and light sensor. The system operates smoothly because the Arduino board and sensor mounting structures and power supply with cables integrate all components properly. The system includes a miscellaneous supply budget to support assembly work and maintenance activities and unexpected needs which will maintain system robustness and full functionality.

Table 5.1: Estimated Budget 1

Tool	Cost (LKR)
Robo Arm	20,000.00
Capacitive Proximity Sensors	6,000.00
Arduino Mega Board	5,000.00
Sensor Mounting Structure	2,500.00
Standard Webcam	3,900.00
USB Webcam (1080p) x 2	11,800.00
USB Microphone	7,500.00
Decibel Meter / Light Sensor	4,500.00
Noise Sensor	3,500.00
Power Supply & Cables	9,500.00
Optional Safety Sensors	8,000.00
Ultrasonic Sensor (HC-SR04)	650.00
Breadboard	300.00
Other Supplies	10,000.00
Estimated Total Budget	93,150.00

6. GANTT CHART

Table 6.1: Gantt Chart 1

No	Assessment (Milestones)	2025 - 2026												
		May (25)	Jun (25)	Jul (25)	Aug (25)	Sep (25)	Oct (25)	Nov (25)	Dec (25)	Jan (26)	Feb (26)	Mar (26)	Apr (26)	May (26)
1	Project Discussion Workshop 1													
1.1	Project Discussion Workshop 2													
2	Topic Evaluation													
2.1	Select a Topic													
2.2	Select a supervisor													
2.3	Topic Evaluation Form Submission													
3	Project Proposal Report													
3.1	Create Project Proposal Report (draft)													
3.2	Create Project Proposal Presentation													
3.3	Project Proposal Presentation													
3.4	Create Project Proposal (Individual)													
3.5	Create Project Proposal (Group)													
4	Develop the System													
4.1	Identifying Functions													
4.2	Database Designing													
4.3	Implementation													
4.4	Unit testing													
4.5	Integration Testing													
5	Progress Presentation - I													
5.1	Project Status document													
5.2	Create Presentation Document													
5.3	Progress Presentation – I (20%)													
6	Research Paper													
6.1	Create the Research Paper													
6.2	Final Research paper													

7	Progress Presentation - II													
7.1	Create Presentation Document													
7.2	Progress presentation – II (80%)													
8	Final Report Submission													
8.1	Final Report Submission													
8.2	Application Assessment													
8.3	Project Status Document													
8.4	Student Logbook													
9	Final Presentation & Viva													
9.1	Create Final Presentation													
9.2	Final Report Submission													

7. CONCLUSION

Collaborative Texture Recognition is an important robotics development that enables autonomous robots to perceive, classify, and respond to objects in terms of surface properties such as roughness, transparency, reflectivity, and material type (e.g., metal, plastic, glass, rubber). By integrating multi-modal sensors like inductive, proximity, optical sensors, and high-resolution cameras with advanced machine learning algorithms like Convolutional Neural Networks (CNNs), transfer learning, and reinforcement learning, robots can achieve high accuracy in recognizing known and unknown textures.

This paper brings to the foreground the importance of human-in-the-loop interaction under which the system is adapted to uncertain or uncertain situations by employing recovery feedback. Robustness is enhanced with multi-modal sensor fusion. Furthermore, real-time data acquisition, preprocessing, and adaptive exploration policies empower the system to act effectively in dynamic and unstructured environments.

The envisioned architecture surmounts some of the most significant challenges in conventional texture recognition systems, namely variability of the environment, limited data, need for real-time processing, and unexplainability. Leverage both vision and touch information, combined with smart learning techniques, the system improves sorting efficiency and accuracy and builds an unobtrusive cooperation between the robot and human.

At a broader level, this module offers a starting point for intelligent, adaptive, and trustworthy robotic sortation machines that can perform advanced industrial and service functions, which will result in wide application of collaborative robots in material identification, industrial automation, and assistive environments.

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